Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 12 - Week 12 (April 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following is a suitable use case for a blockchain-based solution in real-world systems?

a. Storing high-frequency sensor data that requires rapid deletion and modification
b. Creating a transparent and tamper-evident supply-chain tracking system for goods
c. Running a machine learning model training directly on a public blockchain
d. Hosting large video files for a global streaming platform

Answer: (b)

Detailed solution:
Blockchain technology is best suited for applications that require transparency, immutability, and trust among multiple parties. Supply-chain tracking benefits from these properties, as it allows all participants to verify the origin and movement of goods without relying on a central authority.

QUESTION 2

In interbank settlements, what issue does Project Ubin Phase 2 try to solve using decentralized multilateral netting?

a. Preventing double-spending of digital currencies.
b. Verifying identities between banks
c. Resolving gridlock situations where banks lack liquidity for individual payments but are overall solvent
d. Stopping unauthorized access to central bank accounts

Answer: (c)

Detailed solution:
Gridlock happens when banks can not complete payments due to temporary liquidity shortages, even if they are net-positive. Netting allows multiple obligations to cancel out efficiently.

QUESTION 3

The Stellar Consensus Protocol (SCP) uses federated voting. Which of the following are the key steps involved?

a. Propose
b. Vote
c. Accept
d. Prepare
e. Confirm

Answer: (b), (c), and (e)

Detailed solution:
SCP achieves consensus through federated voting among nodes. The simplified key steps involve nodes voting on proposed values, accepting them once sufficient agreement is reached, and finally confirming them to finalize the decision.

Please also refer to week 12 lecture 57.

QUESTION 4

According to research by Amores-Sesar et al., what level of overlap in trusted nodes (UNL overlap) is now recommended for safety in the Ripple network?

a. 20%
b. 40%
c. 80%
d. Over 90%

Answer: (d)

Detailed solution:
A very high overlap (above 90%) ensures that most nodes agree on the same trusted validators, which improves network safety and consistency.

Please also refer to week 12 lecture 57.

QUESTION 5

In the CollabFed decentralized marketplace model, how is a request from a public blockchain verified and transferred to a private consortium ledger?

a. Through a central broker API
b. Using SPV (Simplified Payment Verification) plus approval from two-thirds of service providers
c. By running Proof of Work on the private ledger
d. By encrypting it with a shared private key

Answer: (b)

Detailed solution:
SPV verifies that the request exists on the blockchain, and a supermajority of providers ensures trust before processing it privately.

Please also refer to week 12 lecture 59.

QUESTION 6

Why is it better to use a blockchain-based access log for sensitive government data instead of duplicating the data itself?

a. It speeds up access to large files
b. It allows deletion of stored data
c. It minimizes breach risks while keeping a transparent record of access
d. It removes the need for centralized storage

Answer: (c)

Detailed solution:

Instead of copying sensitive data everywhere, logging access on a blockchain improves transparency and reduces security risks.

Please also refer to week 12 lecture 58.

QUESTION 7

How does CollabFed combine multiple service provider signatures into a single response for users?

a. Schnorr multi-signatures
b. CoSi (Collective Signing) using BLS signatures
c. Ring signatures
d. HMAC

Answer: (b)

Detailed solution:
CoSi with BLS allows multiple signatures to be aggregated into one compact, verifiable signature, improving efficiency.

QUESTION 8

What was the main goal of Project Ubin Phase 3 (Delivery versus Payment)?

a. Automating GST collection
b. Enabling atomic exchange of cash and securities across different systems
c. Supporting cross-border payments
d. Building a retail payment interface

Answer: (b)

Detailed solution:
DvP ensures that asset transfer and payment happen simultaneously, reducing risk in financial transactions.

Please also refer to week 12 lecture 57.

QUESTION 9

What performance characteristics were observed for the Stellar network?

a. Transaction throughput of approximately 350 TPS.
b. Transaction finality/latency of approximately 1 second.
c. Energy consumption comparable to Bitcoin mining.
d. Confirmation times ranging between 3 to 5 minutes.

Answer: (a) and (b)

Detailed solution:
Stellar is designed for fast and efficient transactions, with high throughput and near-instant finality.

QUESTION 10

In decentralized marketplaces without platforms like Booking.com, what major challenge still remains?

a. Difficulty ordering transactions
b. Lack of quick authority for real-time dispute resolution
c. Poor blockchain interoperability
d. High smart contract costs

Answer: (b)

Detailed solution:
Without a central authority, handling disputes (like service issues in real time) becomes much harder.

Please also refer to week 12 lecture 60.